



CSE334s - Database Systems

Gym Management System

Team 2

BN	ID	NAME
103	2300354	KAREN NASSER ABD EL-MALEK
145	2300075	IBRAHIM ALAA IBRAHIM
17	2300178	HABIBA MOHAMED ABD EL-MOHSEN
177	2300509	MARINA GEORGE BOSHRA
33	2300360	LOBNA AHMED ABD EL HAMID
51	2301016	ZEINA WAEEL MAHMOUD
99	2300337	OSAMA AHMED MAHMOUD



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Project Theme

This project implements a relational database system for managing a multi-branch gym organization. The system is built on SQL Server and covers all core operational aspects of a modern gym: member registration and tracking, trainer and class management, membership plans, workout planning, equipment maintenance, and branch administration.

The goal is to provide a structured, consistent, and efficient way to store and retrieve gym data across multiple branches, ensuring data integrity through foreign key constraints, check constraints, and normalized table design.

Forms and Their Functionalities

The application provides a dashboard with five management forms. Each form connects directly to the database and allows users to perform full CRUD operations (Create, Read, Update, Delete).



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Member Management

This form is the primary interface for managing gym members. It displays two data grids: one showing all member records and one showing member phone numbers.

Member Mangment

Back

FITNESS

First Name

Last Name

Gender

Membership

Branch

Plan

Phone

ID

Membership Start Date

May 2026

Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

Today: 5/7/2026

Add Member

Search For Member

Delete Member

Update Member

ID	Fname	Lname	Gender	Join_Date	Membershi	Start Date	Branch_ID	Plan_ID
1	Mariam	Adel	Female	4/1/2026	1	4/2/2026	1	1
2	Youssef	Hany	Male	4/3/2026	2	4/4/2026	2	2
3	Salma	Tarek	Female	4/5/2026	3	4/6/2026	3	3

Member_ID	Phone
1	01211111111
2	01222222222
3	01233333333

Add Phone

Delete Phone

Functionalities:

- Add a new member by filling in First Name, Last Name, Gender, Membership, Branch, Plan, Phone, and selecting a Membership Start Date from the calendar picker.
- Search for a member.
- Update an existing member's information after selecting them from the grid.
- Delete a member record from the system.
- Add Phone or Delete Phone to manage multiple phone numbers per member independently.

Member Management Accesses: *Member*, *Member_Phone*, *Membership*, *Workout_Plan*, *Branch_Manager*

The form reads *Membership*, *Workout_Plan*, and *Branch_Manager* to populate the dropdowns so the user picks valid foreign key values. It writes directly to *Member* and *Member_Phone*. This scope makes sense because a member is the central entity that connects to almost everything — their branch, their plan, and their membership tier all need to be validated at the point of entry.



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Procedures:

1. Constructor Procedure

- public Member()

2. Data Loading Procedures

- LoadMemberData()
- LoadPhoneData(int memberId = -1)
- LoadMembershipOptions()
- ClearFields()
- LoadBranchOptions()
- LoadPlanOptions()
- SetupGenderComboBox()

3. Event Procedures

- Add_Member_Click
- Delete_Member_Click
- Search_Click
- Add_Phone_Click
- btnNoPlan_Click
- comboBox1_SelectedIndexChanged
- dataGridView1_CellClick
- Back_Click



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Trainer Management

This form has two tabs: Manage Trainers and Manage Workout Plans, keeping trainer and plan management in one place.

ID	Fname	Lname	Gender	Shift	Speciality	Phone	Branch_ID
1	Ali	Mahmoud	Male	Morning	Cardio	01111111111	1
2	Nour	Samy	Female	Evening	Yoga	01122222222	2
3	Omar	Khaled	Male	Morning	Strength	01133333333	3

ID	Name	Duration	Intensity Level	Trainer
1	Weight Loss	8	Medium	Ali Mahmoud
2	Muscle Gain	12	High	Nour Samy
3	Fitness Starter	6	Low	Omar Khaled

Manage Trainers tab:

- Add, update, delete, and search trainers.
- Fields include First Name, Last Name, Gender, Shift, Specialty, Phone, and Branch.
- Dropdown menus enforce valid values for Gender, Shift, Specialty, and Branch.
- Search for a trainer.

Manage Workout Plans tab:

- Add and delete workout plans.
- Fields include Plan Name (dropdown), Duration in weeks, Intensity Level (dropdown), and assigned Trainer.
- Displays all plans in a grid showing ID, Name, Duration, Intensity Level, and the full trainer's name.
- Search for a Workout Plans.

Trainer Management Accesses: *Trainer, Workout_Plan, Branch_Manager*

Trainer management reads *Branch_Manager* for the branch dropdown and writes to both *Trainer* and *Workout_Plan*. These two are kept together in one form because a workout plan can't exist without a trainer — it's a direct 1:M dependency, so it made sense to manage them in the same place rather than a separate form.



Gym Management System



Procedures:

1. Constructor Procedures

- public Trainers()

2. Data Loading Procedures

- LoadTrainerData()
- LoadWorkoutPlanData()
- LoadBranchOptions(string gender = "")
- ClearPlanFields()
- ClearFields()

3. Event Procedures

- Add_Trainer_Click
- Delete_Trainer_Click
- Update_Trainer_Click
- Search_Click
- Back_Click
- DataGridView1_CellClick
- Add_Plan_Click
- Delete_Plan_Click
- Search_Plan_Click



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Class Schedule Management

This form manages gym classes and their schedules. It displays two grids side by side: one for classes and one for class schedules.

ID	Name	Trainer
1	Morning Cardio	Ali Mahmoud
2	Yoga Flow	Nour Samy
3	Strength Basics	Omar Khaled

Class_ID	ClassName	Day	Time
1	Morning Car...	Monday	08:00:00
3	Strength Bas...	Friday	09:00:00
2	Yoga Flow	Wednesday	17:00:00

Functionalities:

- Add a new class by entering a Class Name and selecting a Trainer.
- Update an existing class name.
- Delete a class from the system.
- Add a Schedule to a class by selecting the Day (dropdown with all 7 days) and Time.
- Delete Schedule entries from the schedule grid independently.

Class Schedule Management Accesses: *Class*, *Class_Schedule*, *Trainer*

The form reads Trainer to populate the trainer dropdown when adding a class and writes to *Class* and *Class_Schedule*. These two tables are tightly coupled, a schedule entry is meaningless without the class it belongs to, so managing them together on one screen is the best choice.



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Procedures:

1. Constructor Procedures

- ClassScheduler()

2. Data Loading Procedures

- LoadClassData()
- LoadTrainerComboBox()
- LoadScheduleData()
- LoadDayComboBox()

3. Event Procedures

- btnAddClass.Click
- btnDeleteClass.Click
- btnUpdateClass.Click
- dgvSchedules.CellClick
- dgvClasses.CellClick
- btnClear.Click



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Membership Management

This is a straightforward form for managing membership tiers offered by the gym.

The screenshot shows a web application window titled "MemberShip". It features the "FITNESS" logo at the top center. Below the logo, there are three input fields: "ID" (a text box), "Duration" (a dropdown menu), and "Price" (a text box). To the right of these fields is a "Back" button. Below the input fields are four buttons: "Add MemberShip", "Update MemberShip", "Delete MemberShip", and "Search". At the bottom of the window is a table with three columns: "ID", "Duration", and "Price". The table contains three rows of data.

ID	Duration	Price
1	1 Month	300.00
2	3 Months	800.00
3	6 Months	1500.00

Functionalities:

- Add a new membership plan by specifying an ID, Duration (dropdown: 1 Month, 3 Months, 6 Months), and Price.
- Update an existing membership plan's detail.
- Delete a membership tier from the system.
- Search for a specific membership.
- All current membership tiers are displayed in a grid with ID, Duration, and Price.

Membership Management Accesses: *Membership* only

This is the most isolated form by design. *Membership* tiers are reference data, they don't depend on any other table, and other tables (like *Member*) just reference them.



Gym Management System

Procedures:

1. Constructor Procedure

- public Membership()

2. Data Loading Procedures

- LoadMembershipData()
- SetupDurationComboBox()
- ClearFields()
- StyleDataGridView()
- ApplyStyle(DataGridView dgv)

3. Event Procedures

- Add_MemberShip_Click
- Update_MemberShip_Click
- Delete_MemberShip_Click
- Search_Click
- Back_Click
- dataGridView1_CellClick



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Branch Management

This is the most comprehensive form, managing branches, their equipment, and the classes each branch offers.

The screenshot shows a web application window titled "Branch Management". It contains several input fields at the top: "Branch Type" (a dropdown menu), "City", "Area", "First Name", "Last Name", and "Phone". Below these fields are three buttons: "Add", "Update", and "Delete". To the right of these buttons is a "Search" button and a "Back" button. Below the input fields is a table with 8 columns: "Branch_ID", "Type", "City", "Area", "Manager_ID", "Manager_Fname", "Manager_Lname", and "Manager_Phone". The table contains 3 rows of data. Below this table is a table with 6 columns: "Branch_ID", "Class_ID", "ID", "Name", "Purchase_Date", "Maintenance_Du", and "Branch_ID". The table contains 3 rows of data. To the right of this table is a table with 3 columns: "Equipment Name", "Purchase Date", and "Maintenance Duration". The table contains 3 rows of data. Below the equipment table are three buttons: "Add", "Update", and "Delete".

Branch_ID	Type	City	Area	Manager_ID	Manager_Fname	Manager_Lname	Manager_Phone
1	Male	Cairo	Nasr City	1	Ahmed	Ali	01011111111
2	Female	Giza	Dokki	2	Sara	Omar	01022222222
3	Mixed	Alex	Smouha	3	Mona	Hassan	01033333333

Branch_ID	Class_ID	ID	Name	Purchase_Date	Maintenance_Du	Branch_ID
1	1	1	Treadmill	1/10/2024	90	1
2	2	2	Yoga Mat	3/15/2024	180	2
3	3	3	Barbell Set	11/20/2023	120	3

Equipment Name	Purchase Date	Maintenance Duration
	Thursday , May 7, 2026	

Functionalities:

- Add, update, delete, and search branches using the top section of the form.
- Fields include Branch Type (Male / Female / Mixed), City, Area, Manager First Name, Last Name, and Phone.
- The Offers grid at the bottom-left shows which classes each branch offers (Branch_ID, Class_ID).
- Search for a Branch by ID.
- The Equipment grid shows all equipment assigned to branches (ID, Name, Purchase Date, Maintenance Duration, Branch_ID).
- Add, update, and delete equipment records directly from this form using the panel on the right side.

Branch Management Accesses: *Branch_Manager, Equipment, Offers, Class*

This is the broadest form. It writes to *Branch_Manager* and *Equipment*, and reads *Offers* and *Class* to display which classes each branch provides. The reason this form is the widest in scope is that a branch is the physical container for everything - equipment lives in a branch, classes are offered by a branch. It made sense to give branch managers a full operational view of their branch from one screen.



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Procedures:

1. Constructor Procedure

- public Branch(DashBord dash)

2. Data Loading Procedures

- LoadBranchData()
- LoadEquipment(int branchId = -1)
- LoadOffers(int branchId = -1)
- SearchBranchById(int id)

3. Event Procedures

- btnAdd_Click
- btnUpdateEquipment_Click
- btnDelete_Click
- btnDeleteEquipment_Click
- btnSearch_Click
- dgvEquipment_CellClick
- dgvBranch_CellClick
- btnBack_Click
- btnAddEquipment_Click



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UI Flow

